

Level 1

Virtual Reality

TASK1:

Put the words in the right form to complete this definition of VR:

VR is a (reality) ----- three-dimensional image or artificial environment that is created with a (mix) ----- of ----- (interaction) hardware and software, and presented to the user as a real environment in which it is interacted with in a seemingly real or physical way. There are several categories of virtual reality technologies. These various types (difference) ----- in their levels of (immerse) ----- as well as (apply) ----- and use cases.

TASK 2: TYPES OF VIRTUAL REALITY

a) Match the heading to the right paragraph .

A *Non- immersive* **B** *Semi- immersive* **C** *Fully – immersive*

☐ The simulations closely resemble and utilize many of the same technologies found in flight simulation. They are powered by high performance graphical computing systems, which are often then coupled with large screen projector systems or multiple television projection systems to properly stimulate the user's visuals.



☐ Hardware such as head-mounted displays and motion detecting devices are used to stimulate all of a user's senses. The simulations are able to provide very realistic user experiences by delivering a wide field of view, high resolutions, increased update rates (also called refresh rate), and high levels of contrast into a user's head-mounted display (HMD).



☐ Only a subset of the user's senses are stimulated, allowing for peripheral awareness of the reality outside the virtual reality simulation. Users enter into these three-dimensional virtual environments through a portal or window by utilising standard high resolution monitors powered by processing power typically found on conventional desktop workstations.



b) Pick up the differences between the 3 categories of VR

Immersive : adj make users immersed and totally involved in a particular environment

Text

VR is already making great inroads into education, with a large number of startups and established companies offering packaged experiences and services aimed at schools. Engage's platform is used by the likes of Facebook, HTC, and the European Commission to enable remote learning. One study published in 2019 found that medical students trained using VR were able to carry out certain procedures quicker and more accurately than peers trained using traditional methods.

These new methods of teaching and learning will become increasingly effective as new technologies emerge. One that is likely to make waves is the Tesla suit, which uses a full-body suit to offer haptic feedback, enhancing the immersion through the sense of touch. It also offers an array of biometric sensors enabling the user's heartbeat, perspiration, and other stress indicators to be measured. The suit is already used in NASA astronaut training, but its potential uses are unlimited.

For training, it could be used to safely simulate any number of hazardous or stressful conditions and monitor the way we respond to them. For example, Walmart has used it to train retail staff to work in Black Friday situations, instructing them on how to best to operate in busy shop environments with long queues of customers.

As well as training us for dangerous situations, it will also drastically reduce the financial risks involved with letting students and inexperienced recruits loose with expensive tools and machinery in any industry.

READING COMPREHENSION

1-Choose a title for the text

- VR is effective in education at the expense of training.
- VR is useful exclusively in training.
- VR proves advantageous for both education and training.

2- Correct these false statements with details from the text.

A- Only giant tech companies are liable to provide VR offerings in education.

B- Traditional medical training has always proved more reliable than VR based training.

C- VR based training seems significantly costly for the industry.

D- Tesla suit is exclusively used by astronauts.

3- Pick up the 2 sentences which highlight two advantages for VR in training.

A-----
B-----

4- Complete this table to give details about an example of simulated training.

The shop	The simulated situation	Goal of the simulation

5-Find words which mean nearly the same as

*An advance or penetration in sth usually at the expense of sth else: -----

*Not tight or restricted by guidelines: -----

LANGUAGE

1. Study the underlined linkers and fill in the table

While we will always make time for meeting up with friends and loved ones in the real world, as our working and school lives become increasingly remote, it's likely that more of our social interaction will move into the online realm, too. Just as we are no longer barred from careers or educational opportunities due to an increasingly virtualized world, we will have more meaningful ways to connect with other humans as technology improves in this area.

Linker	Meaning	Equivalents
While		
As		
Due to		
To		

2. Correct the underlined language mistakes

Very soon, VR creators will extend sensory hijacking to our other faculties – for example, touch and smell – to deep ----- that sense of immersion. At the same time, the devices we use to visit these virtual worlds will become cheaper and lighter, removing the friction that can currently be a barrier.

I believe Extended reality (XR) – a term that cover ----- virtual reality (VR), augmented reality (AR), and mixed reality (MR) – is going to be ----- one of the most transformative tech trend ----- of the next five years. It will be enabled and augmented by other tech trends, included ----- super-fast networking, that will let us experience VR as a cloud service just like we currently consume music and movies. Moreover artificial intelligence (AI) will provide us of ----- more personalized virtual worlds to explore, even giving us realistic virtual characters to share our experiences with.

3. Put the bracketed words in the right tense or form.

The potential harm caused by VR is well established, with negative side effects that **(be /report)** ----- since the early 1990s. Perhaps the most well-documented and prevalent negative consequence of VR is that it can cause **(use)** ----- to experience vertigo, nausea, or dizziness also referred to as cybersickness or simulator sickness. **(Fortune)** -----, most cybersickness symptoms are mild and subside relatively quickly. More recently, research **(explore)** ----- and demonstrated other negative consequences related to VR, including **(reduce)** ----- **(cognition)** ----- performance (e.g., reaction times, physical fatigue, eye fatigue, **(posture)** ----- **(stable)** ----- and physical discomfort. Despite the abundance of evidence for potential negative consequences related to VR **(use)** -----, these consequences **(relate to / predominantly)** ----- to negative physical and/or physiological outcomes.

WRITING

Task 1:

The battle for the best VR headset has never been more fierce. Headsets like the Oculus Rift pushing up against PlayStation VR, it's a showdown between titans. Write a paragraph to compare between the two headsets.



Oculus Rift



PlayStation VR

Platform	Windows PC	PlayStation 4
Experience	Positional tracking, Standing VR, Seated VR	Positional tracking, Standing VR, Seated VR
Field of view	110 degrees	100 degrees
Display resolution per eye	1080 x 1200	1080 x 960
Headset weight	1.0 lbs	1.3 lbs
fit	Fits glasses, interpupillary distance adjustment	Fits glasses, lens distance from eye can be adjusted, interpupillary distance adjustment
Price	\$ 399	\$ 339
Included in the box :	Oculus Rift Headset, Camera Sensor, Remote, Xbox One Wireless Controller, Xbox USB Wireless Adapter, 2 x AA Batteries, Intergrated Headphones Removal Rool, 2 x Oculus Logo Stickers, Oculus Lens Wipe Cloth	Headset, Processor Unit, Connection Cable, HDMI Cable, USB Cable, Stereo Headphones, AC Power Cord, AC Adapter, Demo Disc

Task 2:

Use the notes of the lesson as well as the internet to write short paragraphs in which you define these VR-related terms:

Haptic devices - Immersion